

A REVIEW ON TECHNIQUES FOR SPEECH TO TEXT CONVERSION AND TEXT SUMMARIZATION

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Abstract: *Speech-to-text and text summarization are two key areas, which are part of automatic natural language processing, which have been very useful in recent times as there has been an upsurge in the amount of digitized information as well as the need for efficient transcription and distilling information into its essence. This review paper evaluates the conventional approaches and structural patterns registrable in the recent innovations across both opuses with a keen interest on their complexities, centrality, and space of application. The first issue addressed is related to the transcription of speech into text. These include deep learning and end to end models like transformer networks for neural network-based speech transcription as well as cloud-based services. Several strengths and weaknesses of these methods are assessed relative to precision, time, and flexibility towards different languages and trends within same languages. Then the paper moves on to examine the text summarization techniques. Two types of text summarization techniques are compared, namely extractive and abstractive text summarization, emphasizing advances by transformer models such as BERT and GPT in generating context sensitive summaries. The last focus of the paper is on the speech recognition and summation technologies in practice as used by virtual assistants, meeting transcription applications, and content generation tools.*

Keywords: *Machine learning, Speech to Text, Speech recognition, communication, Machine translation, Automatic Speech Recognition (ASR), WhisperX.*

INTRODUCTION

The introduction of artificial intelligence (AI) and natural language processing (NLP) tools has revolutionized the way content is consumed. Speech-to-text transformation and text summarization stand out as linguistics technologies that are pivotal to effective interaction and access to information in the present global village. With the emergence of various voice- based, automatic transcription and text saturation over usage, the need for efficient speech-to-text translation and text summarization has grown greatly. The speech recognition technology such as ASR is of great importance since it encodes spoken words and phonemes into textual representation, hence supporting the ability to search and utilize voice data on various platforms. Before, systems ASR were based on statistical approaches of the analysis of the speech signal, such as Hidden Markov Models (HMM) and Gaussian Mixture Models (GMM) – which they used together with the language and acoustic models to derive a reasonable accuracy. Nevertheless, with the development of deep learning and end-to-end neural models, ASR has evolved and now these techniques offer higher accuracy rates and an ability to recognize varying accents, languages, and contexts more effectively. Text summarization uses the information available in long texts to produce an overview that is shorter in terms of content and wording. It can also either be extractive which selects certain key sentences in the original text or be abstractive where a new summary is created that states the main point of the original text. Furthermore, in recent years transformer-based models such as BERT and GPT have successfully performed both extractive and abstractive types of text summarization achieving new heights in coherence and target relevance. This review paper aims to provide a comprehensive overview of current speech-to-text and text summarization methods, examining their underlying technologies, strengths, limitations, and real-world applications. By exploring these interrelated fields, we highlight the latest innovations and identify future directions to improve the integration of these technologies in diverse industries.

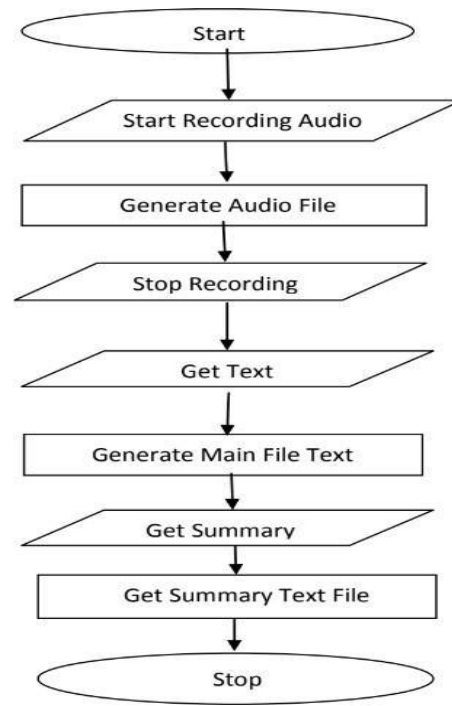


Figure 1 High level overview of Project

Methods :

The methods for speech to text conversion and text summarization can be divided into categories such as cloud based, Neural Network based and Transformer based. Using the combination of these approaches multiple architectures are created. Some of these architectures are PodSumm architecture [11] and BertSumm (transformer) architecture

1.1 Cloud based

The speech is transformed into text by cloud hosted advanced speech-to-text technologies by using sophisticated machine learning models. Alike Google Cloud Speech-to-Text, Amazon Transcribe, and Microsoft Azure Speech Service offer integration and customization for transcription services, automated punctuation, identification of call participants, and support for multiple languages, both real-time and in bulk. They provide accurate services due to the presence of multi deep learning models on these platforms. Additionally, cloud-based services eliminate the need for local infrastructure, offering accessibility from anywhere, easy integration into applications, and the ability to handle large volumes of audio data efficiently.

Though the cloud-based methods are effective over the time for the production use the cost of the services may add up and cost heavily with increase in the number of the consumers. Cloud based methods are best used for a small-scale project. These methods are easy to setup and implement due to them being heavily abstracted.

1.2 Neural Network

Long Short-Term Memory (LSTMs)

The paper [2][9] dives into the design and usefulness of LSTM systems, emphasizing their capacity to capture successive conditions in content information Through an examination of context- based text generation techniques, the author showcases the efficacy of LSTM networks in generating coherent and contextually relevant textual content. text generation models, highlighting their significance in tasks such as machine translation, dialogue systems, and summarization. Furthermore, the author presents experimental results and performance evaluations to demonstrate the effectiveness of LSTM networks in context-based text generation, paving the way for advancements in natural language processing and artificial intelligence research." Through a detailed examination of context-based text generation techniques, the author demonstrates the effectiveness of LSTM networks in generating coherent and contextually relevant textual

output. The document outlines methods of incorporating contextual information into LSTM-based models, illustrating their use in machine translation, dialogue generation, summarization, and other fields.

Recurrent Neural Networks (RNN)

The paper [1] and [3] present us with structure of RNN network and its various implementations such as RNN Transducer. RNNs are ideal in tasks such as speech to text and text summarization RNNs have a unique perspective on speech because they continuously strive to convert audio features into text sequences. This enables them to transform audio features into textual forms. Such an ability is advantageous as it allows a network to understand context in the spoken language. RNNs are therefore able to compensate for varying patterns of speech and pronunciation accents that may exist across different individuals. These transcribers are more effective for RNNs' greater transcription success increases their effectiveness in a broad range of applications including voice- command mechanics and automated transcription systems.

Deep RNNs also consists of various layers such as Connectionist Temporal Classification, RNN transducer which are decoded by a decoding function and its output is regularised using Regularization process. The paper experiment on various combinations of the layer to find the most optimal and low error producing layer.

Extractive Summarization

The authors of paper utilizes Extractive summarization to generate a text summary of the input text. The words in the text are usually split and made ready for processing as the first step. The next step is feature engineering, where relevant qualities are chosen, like creating a feature representing a word using word embeddings which encode words semantically in numerical forms. After this, a RNN, an LSTM network, or any other neural network model is implemented and trained on labeled data to understand how sentences correspondence to their context and how important they are. To sum up, the model processes every sentence independently and assigns each a score indicating how important it is in the context of the paragraph. After that, the scores are used to filter the sentences, and the highest scoring sentences are selected and put together into a summary that is not only clear but also comprises significant information from the original text. Finally, the selected sentences are organized and formatted to produce the final summary [4][8].

1.3 Transformer based

Generative Adversarial Networks (GANs)

The overview [2] comprehensively explores the application of GANs in creating literary substance, highlighting their potential over different spaces. The GANs are built over the top of transformers [6]. By analyzing the basic standards and designs of GANs in content era, the overview illustrates the points of interest and challenges related with this approach. Assessment strategies particular to GAN-based content era, along with relevant evaluation measurements, are examined to evaluate the quality and coherence of created content. Moreover, the study addresses eminent progressions, rising patterns, and future investigate bearings in leveraging GANs for text generation, underscoring their importance in progressing the field of common language preparing. The overview fastidiously analyzes the utilization of GANs over a range of text generation assignments, counting language modeling, dialogue generation, and story generation. By dissecting the engineering and preparing methods of GANs in text generation. The study explains the challenges and openings characteristic in this approach.

Two-Stage Transformer

In the paper [5], the authors proposed and created a two-stage approach using the transformers [6] for variable-length summary has been proposed. Initially, the text segmentation module makes use of a pre-trained BERT and a bidirectional LSTM network to segment the input text. Next, BERTSUM or the BERT based extractive summarization model is constructed in order to perform the essential task of

extracting the most salient sentence from each segment. With respect to training the two-stage summation model, the first step would involve employing the extracted sentences to train or language build the document summarization model depicted in the second stage. Thereafter, the extracted segments are used to train the specific summarisation module depicted in the first stage with the consideration of the outputs of this module and the documents summarisation module from the second stage. The parameters defining the segment summarization module are perturbed based upon the cumulative loss scores of the document summarization and segment summarisation modules. After this stage of training, an alternate training method termed as collaborative training is performed in which the parameters defining the segment summarisation and documents summarisation modules are trained repeatedly until the parameters converge. Integrating the outputs of the segment summarization module To test the framework the outputs of the segment summarization module are combined, to generate variable length one-shot abstractive summary. To be compared with, the BERT-biLSTM- based text segmentation model is tested on the ChWiki_181k database and yields a satisfactory impact on seizing the connection between sentences. Lastly, the proposed variable-length abstractive summarization model attained the maximum accuracy of 70.0% on the LCSTS dataset (in human subjective evaluation).

Dual-encoding using transformer

Attentional models that are based on recurrent neural networks and sequence to sequence are shown to be effective in abstractive text summarization [8]. The author in the paper [7] has suggested a framework of abstractive text summarization casted in a dual encoding. The suggested technique involves the use of two encoders i.e., the primary and the secondary encoder. Precisely, the first encoder performs rough encoding in a normal fashion and the second encoder approximates the meaningfulness of the words and gives more finer encoding with the input raw text and the output text-encoding summarization accounted for earlier. These two-level encodings are concatenated and sent into the decoder to create a more diverse summary that will reduce repetition phenomenon in generating long sequences. The experimental outcomes of two difficult datasets (i.e. CNN/ DailyMail and DUC 2004) show that our dual encoding model compares with the existing methods.

1.4 PodSumm Architecture

The first step is represented by PodSumm converting the audio podcast to a text transcript with the help of an ASR module and breaking the text transcript into separate sentences [10]. It filters the text summary by using a text summarization model that identifies relevant sentences, their time stamp in the audio, and finally generates the final audio summary whose text summary is. The stages are mentioned below in detail.

1) Automatic Speech Recognition: ASR techniques achieve the process of automatic speech to text translation. Since the objective of this work is neither creation of a new ASR system nor enhancement of some existing one, we desire to solve this particular task with a commonly-known and publicly-available solution in that matter, i.e., AWS Transcribe.

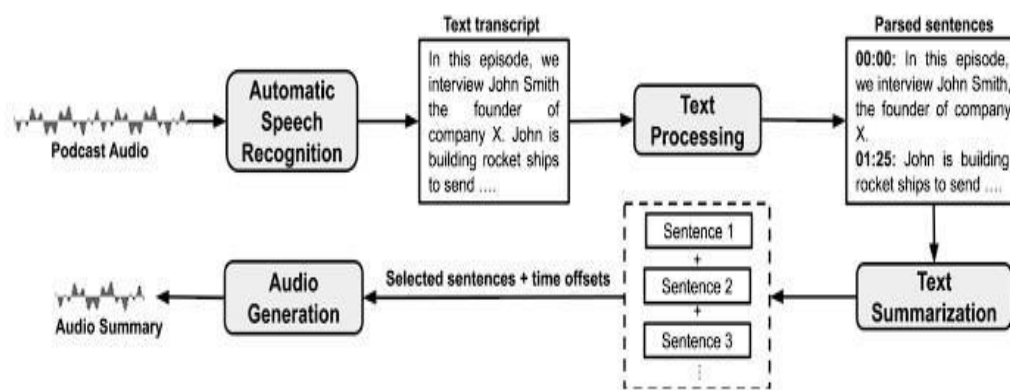


Figure 2 PodSumm Architecture

2) Text Processing: The text associated with these individual words and punctuation marks in the transcripts of ASR module includes the time of occurrence in the audio as well as starting and ending time of the word or punctuation mark and the confidence level of the prediction. The author prefers to apply an open-source NLP library, i.e. spaCy3, to break the text into separate sentences and provide each of them with the start and the end time [13][15]. Also, they require a sentence to be broken in case of a pause longer than two seconds between words.

3) Text Summarization: Automatic extractive summarization assists in selecting the related sentences and then creating a text summary of the transcript. They exploited a recently postulated Pre- Summ4 model [10] that considers BERT [10] to acquire sentence level encoding and layers inter- sentence Transformer to hold the document- level features within summarization. The best result is emphasized in the final trained model (ROUGE-L F- score 0.64) as compared to the baseline method (ROUGE-L F- score 0.52).

RELATED WORK

2.1 Speech to Text conversion

In the following table, a comparative discussion is given about various approaches in speech to text and we have discussed major landmark achieved in this field as well as significant advantages and disadvantages to each approach. For example Hidden Markov Models (HMMs) and Guassian Mixture Models [12] have history with earlier days of speech recognition, where automatic speech recognition was borned outta these statistical models which lead to good useful work done on low resourced data sets etc but NOT in real time. The need for more precise and flexible models has led to the increased use of deep learning-based models in projects such as Recurrent Neural Networks(RNNs) and Long Short Memory(LSTM), which are used since they can handle sequences without losing sight of dependencies[16].

Table 1 Various Speech to Text Conversion Methods

Technique	Description	Advantages	Limitations	Examples/Applications
Hidden Markov Model (HMM)	Statistical model that uses probabilities to predict sequences of sounds to words.	Well-established, simple, and interpretable.	Struggles with noisy environments and diverse accents.	Early ASR systems, dictation software.
Gaussian Mixture Model (GMM)	Used with HMMs to model different sound distributions.	Improved acoustic modeling over pure HMMs.	Sensitive to variations in accents, noise, and requires large datasets.	Used in early versions of Siri, Google ASR.

Recurrent Neural Networks (RNN)	Neural network that processes sequences, making it suitable for speech.	Better at capturing sequential dependencies than statistical models.	Struggles with long-range dependencies and complex speech patterns.	Real-time speech recognition, language translation.
Long Short-Term Memory (LSTM)	A type of RNN designed to retain information over long sequences.	Handles long-range dependencies well, robust in noisy environments.	Computationally intensive, difficult to scale to large datasets.	Used in real-time transcription, virtual assistants.
End-to-End Models	Models like transformers that directly convert speech to text without intermediate steps.	Simplifies pipeline, high accuracy, adaptable to multiple languages.	Requires large datasets for training, may struggle with low-resource languages.	Wav2Vec, Deep Speech, automated transcription services.
Transformer-Based Models	Deep learning models that process entire speech sequences in parallel.	High accuracy, faster processing, adaptable to various accents and dialects.	Complex to train, resource intensive, large data requirements.	Wav2Vec 2.0, Google's Speech-to-Text API.
Cloud based Models	Cloud based speech to text model provided by various	Achieves high accuracy in transcribing audio into text, making it suitable for various applications.	Limited customization options are available; also relies on a stable internet connection for optimal performance.	Google Cloud Speech-to-Text, Amazon Transcribe, Microsoft Azure Speech Service, IBM Watson Speech to Text.
WhisperX	A speech recognition model based on OpenAI's Whisper includes speaker diarization using pyannote.audio.	-High transcription accuracy. -Speaker diarization support. -Multilingual support. -Time-stamped output.	- Requires GPU for faster processing - Larger models can be resource intensive	Multi-speaker meeting transcription, summarization of voice notes, diarized dialogue systems

2.2 Text summarization

The comparative analysis of text summarization technologies presented in the table reveals a diverse array of approaches, each with its unique strengths and limitations. The table below list the most common used algorithms for text summarization.

Table 2 Methods for Text Summarization

Year	Algorithm	Key Developments	Pros	Cons
2016	Rule-based Systems	Utilization of linguistic rules and heuristics for text generation.	Offers transparency and interpretability in the generation process.	Scalability is limited; crafting rules manually can be labor- intensive.
2017	Hidden Markov Models	Application in both speech recognition and generative tasks.	Established approach with a wide range of applications in text modeling.	Struggles to capture complex dependencies between words or phrases.
2018	Recurrent Neural Networks (RNN)	Enhanced mechanisms for improved context modeling, specifically for sequential data.	Effectively manages and processes sequential information, making it suitable for text generation.	Prone to issues like gradient vanishing; slow convergence during training can hinder performance.
2020	Generative Adversarial Networks (GANs)	Introduction of improved training techniques to enhance model stability.	Capable of producing diverse and realistic text samples that mimic human-like generation.	Often faces challenges related to training instability and mode collapse, which can affect quality.
2021	Long Short-Term Memory Networks (LSTM)	Advanced architectures focused on capturing sequential dependencies in text data.	Effectively retains context over longer sequences, improving performance in tasks requiring understanding of continuity.	May still struggle with very long-term dependencies, leading to loss of information over time.
2022	Facebook's BART Model	Utilization of a transformer-based architecture specifically designed for summarization tasks.	Produces coherent and concise summaries, enhancing readability and comprehension.	Requires significant computational resources and can be complex to train effectively.

2019	BERT (Extractive)	Introduction of deep bidirectional transformers for sentence-level embeddings in summarization tasks	Captures rich contextual representations. It is highly effective in extractive summarization and requires no retraining.	Limited to sentence selection only; cannot paraphrase or generate new text; performance depends on quality of input segmentation.
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PROPOSED METHOD

3.1 Audio Preprocessing

3.1.1 Noise Reduction

To make audio input clearer, we use noise reduction with the pydub and noisereduce libraries. The pydub module loads and slices audio files. Noisereduce uses spectral gating techniques to cut down on unwanted background sounds like wind, traffic, or crowd noise. This preprocessing step improves the quality of speech input and increases the accuracy of the transcription model.

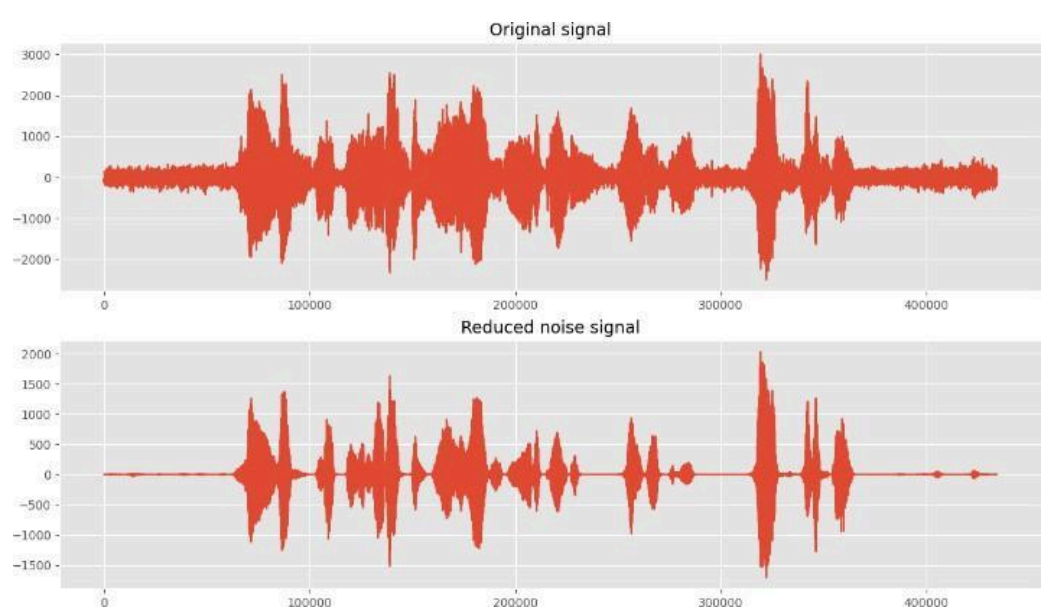


Figure 3 : Original Signal vs Reduced Noise Signal

3.1.2 Justification for Noise Reduction Techniques

Reducing environmental noise is crucial in speech-to-text systems, especially for real-world recordings. Spectral gating suppresses low-intensity frequencies linked to noise while keeping speech elements intact. This process improves the signal-to-noise ratio and enhances overall transcription performance.

3.2 Audio Segmentation

To manage long audio recordings efficiently, we break the audio into smaller pieces using pydub. This process ensures we use memory wisely and keeps accuracy during transcription. It also helps separate speaker turns and long pauses, which makes diarization and summarization easier.

3.3 Speech Recognition and Diarization

Speech recognition uses WhisperX, a newer version of OpenAI's Whisper model. WhisperX supports multiple speakers and produces accurate transcriptions with matched timestamps.[17]

WhisperX incorporates pyannote.audio for speaker diarization, which separates and labels different speakers in the audio. This helps create transcripts that clearly show "who spoke when." This information is essential for meaningful summaries and understanding dialogue.

```
result
{
  'segments': [
    {
      'start': 0.309,
      'end': 3.39,
      'text': ' And RLHF is how we take some human feedback.',
      'words': [
        {
          'word': 'And', 'start': 0.309, 'end': 0.389, 'score': 0.985},
        {
          'word': 'RLHF', 'start': 0.569, 'end': 1.149, 'score': 0.652},
        {
          'word': 'is', 'start': 1.329, 'end': 1.409, 'score': 0.508},
        {
          'word': 'how', 'start': 1.469, 'end': 1.729, 'score': 0.845},
        {
          'word': 'we', 'start': 1.929, 'end': 2.09, 'score': 0.869},
        {
          'word': 'take', 'start': 2.17, 'end': 2.41, 'score': 0.833},
        {
          'word': 'some', 'start': 2.45, 'end': 2.61, 'score': 0.752},
        {
          'word': 'human', 'start': 2.65, 'end': 2.91, 'score': 0.912},
        {
          'word': 'feedback.', 'start': 2.95, 'end': 3.39, 'score': 0.808}],
      'start': 3.97,
      'end': 14.432,
      'text': 'The simplest version of this is show two outputs, ask which one is better than the other, which one the human raters prefer, and then feed that back into the model with reinforcement learning.',
      'words': [
        {
          'word': 'The', 'start': 3.97, 'end': 4.09, 'score': 0.883},
        {
          'word': 'simplest', 'start': 4.33, 'end': 4.67, 'score': 0.899},
        {
          'word': 'version', 'start': 4.71, 'end': 4.99, 'score': 0.86},
        {
          'word': 'of', 'start': 5.03, 'end': 5.07, 'score': 0.908},
        {
          'word': 'this', 'start': 5.11, 'end': 5.27, 'score': 0.869},
        {
          'word': 'is', 'start': 5.35, 'end': 5.45, 'score': 0.557},
        {
          'word': 'show', 'start': 5.61, 'end': 5.81, 'score': 0.855},
        {
          'word': 'two', 'start': 5.85, 'end': 5.99, 'score': 0.851},
        {
          'word': 'outputs', 'start': 6.07, 'end': 6.431, 'score': 0.894},
        {
          'word': 'ask', 'start': 7.171, 'end': 7.431, 'score': 0.823},
        {
          'word': 'which', 'start': 7.531, 'end': 7.691, 'score': 0.814},
        {
          'word': 'one', 'start': 7.771, 'end': 7.851, 'score': 0.907},
        {
          'word': 'is', 'start': 7.911, 'end': 7.971, 'score': 0.75},
        {
          'word': 'better', 'start': 7.991, 'end': 8.211, 'score': 0.834},
        {
          'word': 'than', 'start': 8.231, 'end': 8.351, 'score': 0.877},
        {
          'word': 'the', 'start': 8.371, 'end': 8.451, 'score': 0.837},
        {
          'word': 'other', 'start': 8.511, 'end': 8.691, 'score': 0.714},
        {
          'word': 'which', 'start': 9.571, 'end': 9.691, 'score': 0.993},
        {
          'word': 'one', 'start': 9.751, 'end': 9.811, 'score': 0.866},
        {
          'word': 'the', 'start': 9.831, 'end': 9.891, 'score': 0.998},
        {
          'word': 'human', 'start': 9.931, 'end': 10.151, 'score': 0.883},
        {
          'word': 'raters', 'start': 10.592, 'end': 10.872, 'score': 0.803},
        {
          'word': 'prefer', 'start': 10.932, 'end': 11.292, 'score': 0.872},
        {
          'word': 'and', 'start': 11.932, 'end': 12.032, 'score': 0.691},
        {
          'word': 'then', 'start': 12.072, 'end': 12.252, 'score': 0.812},
        {
          'word': 'feed', 'start': 12.372, 'end': 12.552, 'score': 0.764},
        {
          'word': 'that', 'start': 12.572, 'end': 12.712, 'score': 0.786},
        {
          'word': 'back', 'start': 12.732, 'end': 12.912, 'score': 0.918},
        {
          'word': 'into', 'start': 12.932, 'end': 13.072, 'score': 0.735},
        {
          'word': 'the', 'start': 13.092, 'end': 13.152, 'score': 0.993}],
    }
  ]
}
```

Figure 4 : Diarization of Audio file

[] diarize_segments					
	segment	label	speaker	start	end
0	[00:00:00.008 -> 00:00:00.025]	A	SPEAKER_01	0.008489	0.025467
1	[00:00:00.195 -> 00:00:36.697]	B	SPEAKER_01	0.195246	36.697793
2	[00:00:37.835 -> 00:00:38.327]	C	SPEAKER_01	37.835314	38.327674
3	[00:00:39.159 -> 00:00:44.643]	D	SPEAKER_01	39.159593	44.643463
4	[00:00:45.475 -> 00:00:48.022]	E	SPEAKER_01	45.475382	48.022071
...	...	...	...	...	...
83	[00:07:39.142 -> 00:07:45.526]	CF	SPEAKER_01	459.142615	465.526316
84	[00:07:46.986 -> 00:07:50.161]	CG	SPEAKER_01	466.986418	470.161290
85	[00:07:50.738 -> 00:07:51.061]	CH	SPEAKER_00	470.738540	471.061121
86	[00:07:51.061 -> 00:07:53.471]	CI	SPEAKER_01	471.061121	473.471986
87	[00:07:53.777 -> 00:07:58.463]	CJ	SPEAKER_01	473.777589	478.463497

88 rows x 5 columns

Figure 5 : Timestamps of speakers in Audio file

3.4 Text Preprocessing

Before summarization, the transcribed text is cleaned to improve the input quality for the summarization models. This involves two key steps:

- Expanding contractions such as “I’m” to “I am” or “don’t” to “do not” to ensure consistency across tokens.

- Removing filler phrases like “hello sir,” “thank you,” and “good morning,” which do not add to the meaning but often appear in real conversations.

These preprocessing steps help the summarization models focus on meaningful content.

3.5 Text Summarization

Our system uses a hybrid summarization strategy. It combines extractive and abstractive techniques to meet different summary needs.

3.5.1 Line-by-Line Summary using BERT (Extractive)

To create detailed summaries by speaker, we use the BERT-based extractive summarizer. This model scores each sentence from the transcript according to its context and picks the most informative ones. The outcome is a summary that follows the original wording and order. This makes it useful for things like meeting minutes or breakdowns of conversations.

```
Start Time: 11:07:23

SPEAKER_00: hello sir, good morning. could you please tell us a bit more about the new project we will be working on?
Summary: could you please tell us a bit more about the new project we will be working on?

SPEAKER_02: yes, of course. so, this upcoming project is a fully functional application that we are developing for one of our key foreign clients, zoya enterprises. as you may know, we had some great collaboration with them in the past. this new app will focus on addressing their shipment and logistic challenges.
Summary: so, this upcoming project is a fully functional application that we are developing for one of our key foreign clients, zoya enterprises. this new app will focus on addressing their shipment and logistic challenges.

SPEAKER_00: that sounds amazing. how are we planning to move forward with the development process?
Summary: that sounds amazing. how are we planning to move forward with the development process?

SPEAKER_02: great question. we will be following an industry standard 5 stage development approach to make sure everything runs smoothly and on schedule. so the first step is planning. this is where everything begins. we will sit with the client to get, gather detailed requirements, set project goals, define milestones and develop a timeline.
Summary: we will be following an industry standard 5 stage development approach to make sure everything runs smoothly and on schedule. we will sit with the client to get, gather detailed requirements, set project goals, define milestones and develop a timeline.

SPEAKER_00: got it. that sounds like a solid and well structured plan. i am really looking forward to contributing and learning through this project. it seems like a great opportunity for all of us.
Summary: that sounds like a solid and well structured plan. i am really looking forward to contributing and learning through this project.

SPEAKER_01: thanks for the breakdown. on the devops side, are we planning to use ci/cd pipeline from the beginning? and will infrastructure will be managed with something like terraform or aws cloudformation?
Summary: on the devops side, are we planning to use ci/cd pipeline from the beginning? and will infrastructure will be managed with something like terraform or aws cloudformation?

SPEAKER_02: great point. yes, we will implement ci-cd from the early stages using github actions. as for infrastructure, we are learning towards using terraform for its modularity and cross-cloud capabilities, though we will finalize that in the planning stage.
Summary: yes, we will implement ci-cd from the early stages using github actions. as for infrastructure, we are learning towards using terraform for its modularity and cross-cloud capabilities, though we will finalize that in the planning stage.

SPEAKER_01: phase. perfect. i would love to contribute to setting up the pipeline and managing the infrastructure as code. this project is going to be great learning experience. thank you.
Summary: i would love to contribute to setting up the pipeline and managing the infrastructure as code. this project is going to be great learning experience. .
```

Figure 6 : Summary using BERT

3.5.2 Overall Summary using BART (Abstractive)

To create a brief, high-level summary of the entire transcript, we use BART, a transformer-based summarization model. BART has an encoder-decoder structure that rewrites the input text into a clear and smooth format. Unlike extractive methods, BART generates new sentences by rephrasing the input while keeping the main points. This makes it perfect for summary reports or executive overviews.

3.6 Summary of Approach

- Audio is cleaned and divided for efficient processing.
- Transcription and speaker diarization are done using WhisperX with integrated pyannote [17].
- The text is cleaned by removing contractions and filler phrases.
- A BERT-based model creates a line-by-line extractive summary, keeping speaker turns.
- A BART-based model provides an abstractive overall summary, which is ideal for quick reading.[14]

Architecture Diagram and System Overview

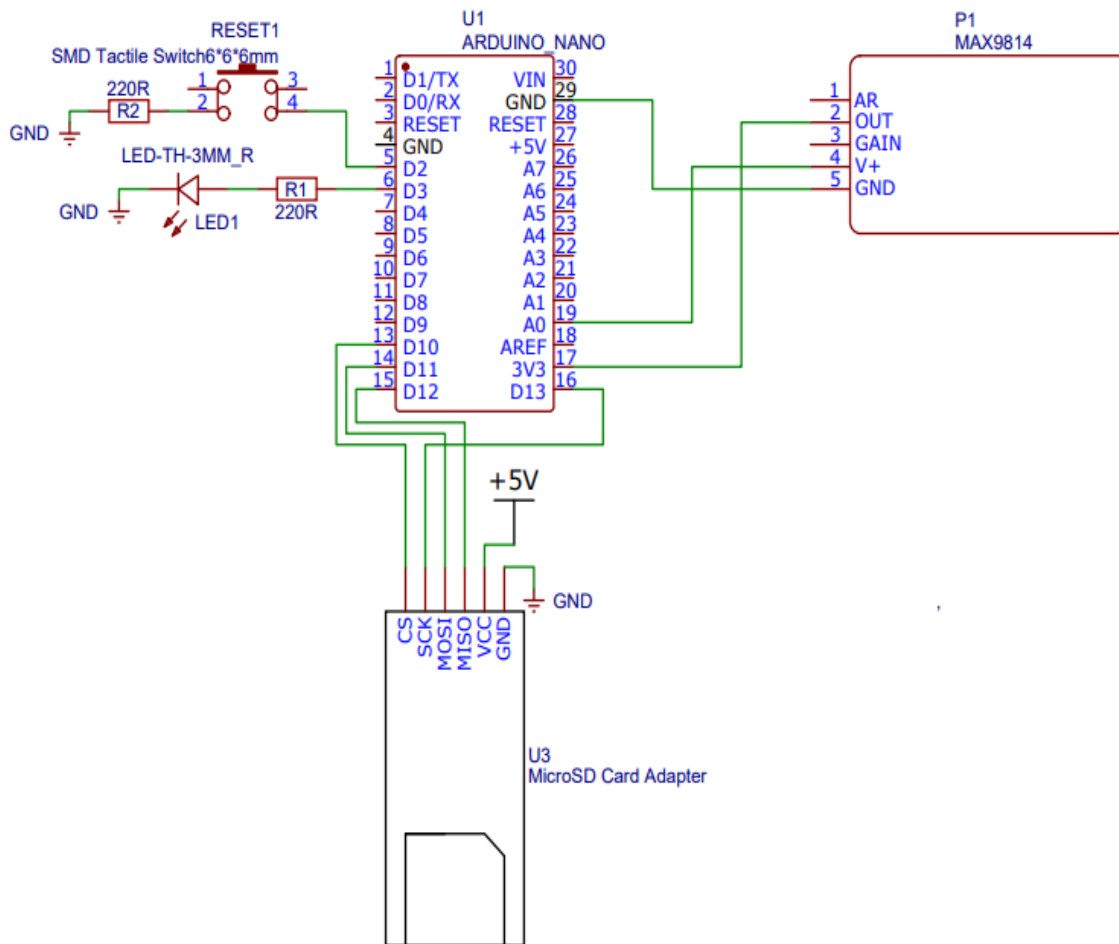


Figure 7 Hardware Circuit Diagram

The proposed system integrates speech-to-text transcription and text summarization using an embedded hardware setup combined with software-driven natural language processing (NLP). The system architecture consists of an audio input module, microcontroller processing unit, storage, and power management components. The figure below represents the architecture of the hardware design used in this study.

Hardware Components and Description

1. MAX9814 Electret Microphone Module

- This is the primary audio capture device used to record speech input. It features automatic gain control (AGC) to optimize audio quality for speech processing.

2. Arduino Nano (ATmega328P)

- The microcontroller is responsible for managing data acquisition, pre-processing signals from the microphone, and controlling peripheral components like storage and RTC.

3. DS1302 Real-Time Clock (RTC) Module + CR2032 Battery

- The RTC module provides accurate timestamping for recorded speech data, ensuring the logs are synchronized for further processing.

4. **SD Card Reader Module (SPI-based) & Micro SD Card (8GB - 32GB, FAT32)**

- The system stores captured speech data and processed text outputs on an SD card for later retrieval and analysis.

5. **USB Mini B Cable & USB Power Bank (2000mAh - 10,000mAh)**

- The microcontroller and peripherals require a steady power source, which is provided via a USB power bank.

6. **Breadboard & Jumper Wires**

- Used for circuit prototyping and establishing connections between various hardware components.

7. **Tactile Push Buttons (3x)**

- These buttons are used for user interaction, such as starting and stopping recordings or triggering summarization functions.

8. **LEDs (Red/Green/Blue, 5mm) – 4**

- Status indicators for system processes like recording, processing, and completion of transcription/summarization.

9. **Resistors (220Ω - 6, 10kΩ - 6)**

- These resistors are used for current regulation in the circuit, ensuring stable operation of LEDs and buttons.

10. **Capacitors (0.1μF (104) Ceramic Capacitor - 2)**

- These capacitors help in power stabilization and noise filtering in the circuit.

Functionality and Workflow

The MAX9814 microphone picks up audio which is then processed by the Arduino Nano. The audio is stored on the SD card so it can be transcribed at a later date. Once the speech-to-text conversion is achieved with modern NLP techniques, the text goes through a multi-step summarization process which includes both extractive and abstractive techniques. This final processed output can be retrieved and used for practical applications such as transcription of meetings, virtual assistants, and content summarization tools.

This is the initial attempt at incorporating speech recognition with summarisation to embedded systems in order to facilitate automatic transcription processes.

RESULTS

Measures aimed to reduce noise contributed a positive improvement as far as transcription clarity was concerned. Moreover, whereas organic approaches to speech canonicalization tended to mute critical spectral constituents, the spectral gating confirmed such crucial bands to provide STT with a pristine input signal. The authors also noted that segmentation was most useful in processing lengthy recordings for there was minimal strain on the memory. This step was necessary to guarantee WhisperX worked with manageable amount of audio segments plus did not slow down WhisperX proved to provide very accurate transcriptions and to display the ability to recognize different speakers in complex environments. Diarization improved the usability of the transcriptions because it added the level of speaker-customary to the transcriptions. The Audio used for the testing had two speakers and the diarization model successfully identified the two speakers along with the duration they spoke for. Table 3 shows the result of the diarization.

Table 3. Diarization and Speech to text results for 30 seconds Audio

Speakers	Start (in seconds)	End (in seconds)	Text
SPEAKER_00	1.6553480475382003	8.395585738539898	Mark? Hi! It's been ages since I last saw you. How are you and Jackie?
SPEAKER_01	8.463497453310698	9.617996604414262	Yeah, good thanks.
SPEAKER_00	10.00848896434635	12.419354838709678	And your new baby? George, isn't it?
SPEAKER_01	12.843803056027166	18.3616298811545	You've got a good memory. Yes, he's two now. What about you? Are you still working in the health centre?
SPEAKER_00	18.972835314091682	27.258064516129032	Yes, for the time being, but we're moving in a couple of months. Anyhow, I better go, I'm late for work. Lovely to see you again.
SPEAKER_01	27.818336162988118	29.83870967741936	Yeah, likewise. Keep in touch

The diarization and speech-to-text conversion results presented in Table 4 were achieved using low-error-rate models on a 5-minute audio file.

Table 4. Error rates for 5 minutes audio

Error Rates	Percentages
Word Error Rate (WER)	9.179 %
Character Error Rate (CER)	3.147 %

CONCLUSION

This paper has proposed an end-to-end framework which combines the spoken transcription process with the extractive and abstractive summary steps without any manual interaction. The system is equipped with a powerful pipeline of transferring speech into brief, meaningful and textual sentences out of the use of WhisperX- a time-critical and speaker-conscious scheme of translation, and enhanced transformer-based techniques Like BERT and BART that are top of the line.

The accuracy of the transcription part proved to be very high, and the word error rate (WER) was low and speaker diarization was also effective. This increased the clarity of written dialogue, especially in multi-speaker audio contexts, e.g. interviews, meetings, or podcasts. What is more is that the two-level summarization (BERT and BART were used to generate line-by-line extractive summaries and abstractive summaries, respectively), also enabled the system to generate contextually rich summaries that could be used in various applications such as documentation and content generation.

Among the main contributions of the work, the practical implementation of state-of-the-art transformer models in a modular system that guarantees performance but promotes adaptability to different types of data inputs and real-world audio data should also be pointed out. The hybrid design of the system provides a valid alternative between academic models or deployment as the system provides a plausible system where automatic transcription and summarization can be implemented in fields of education, healthcare, customer service and journalism.

Future development will be setting up the system to better handle real time processing on low or does not require energy-intensive computing devices as required in edge computing applications. Moreover, by increasing the multilingual support of the system, it will have a broader use on the global bases of linguistic diverse users. These areas strive to address will make it easier to enhance the accessibility, scalability and practical implementation of intelligent speech-to-summary systems in practical setting.

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